Screenshots

1. Add Student

```
--- Student Management ---
1. Add Student
2. Display Students
3. Delete Student
4. Update Student
5. Search Student
6. Sort Students
7. Course Statistics
8. Course Grade Report
9. Course Management
10. Professor Management
11. Login System
12. Exit
Enter choice: 1
Enter Email: srujana.tilli@sjsu.edu
Enter First Name: Srujana
Enter Last Name: Tilli
Enter Course ID: 1111
Enter Grade: A
Enter Marks: 92
Student added successfully!
```

This functionality allows the user to add a new student record by entering details such as email, name, course ID, grade, and marks. The record is stored in the students.csv file using the CSVHandler.

2. Display Students

```
--- Student Management ---
1. Add Student
2. Display Students
3. Delete Student
4. Update Student
5. Search Student
6. Sort Students
7. Course Statistics
8. Course Grade Report
9. Course Management
10. Professor Management
11. Login System
12. Exit
Enter choice: 2
Email: divija.puli@sjsu.edu, Name: Divija Puli, Course: 1111, Grade: A, Marks: 100
Email: sriya.anna@sjsu.edu, Name: Sriya Anna, Course: 1111, Grade: A, Marks: 98
Email: vats.sri@sjsu.edu, Name: Vats sri, Course: 1111, Grade: B, Marks: 88
Email: Mady.dande@sjsu.edu, Name: Mady Dande, Course: 1111, Grade: C, Marks: 78
Email: prad.mohan@sjsu.edu, Name: Prad Mohan, Course: 1111, Grade: B, Marks: 84
Email: srujana.tilli@sjsu.edu, Name: Srujana Tilli, Course: 1111, Grade: A, Marks: 92
```

This feature retrieves all student records from the CSV file and displays them in a readable format. It confirms that data is being loaded correctly from persistent storage.

3. Delete Student

```
--- Student Management ---
1. Add Student
2. Display Students
3. Delete Student
4. Update Student
5. Search Student
6. Sort Students
7. Course Statistics
8. Course Grade Report
9. Course Management
10. Professor Management
11. Login System
12. Exit
Enter choice: 3
Enter Email of student to delete: srujana.tilli@sjsu.edu
Student deleted successfully!
```

This feature removes a student record based on the email ID. It ensures that only valid records are deleted and updates the CSV file accordingly.

4. Update Student

```
--- Student Management ---
1. Add Student
2. Display Students
3. Delete Student
4. Update Student
5. Search Student
6. Sort Students
7. Course Statistics
8. Course Grade Report
9. Course Management
10. Professor Management
11. Login System
12. Exit
Enter choice: 4
Enter Email of student to update: prad.mohan@sjsu.edu
Leave blank to keep current value.
First Name (Prad): Prad
Last Name (Mohan):
Course ID (1111):
Grade (B): C
Marks (84): 75
Student updated successfully!
```

This functionality allows modification of an existing student's information. The user can update selected fields while leaving others unchanged. The updated data is saved back to the CSV file.

5. Search Student (WITH TIME)

```
5. Search Student
6. Sort Students
7. Course Statistics
8. Course Grade Report
9. Course Management
10. Professor Management
11. Login System
12. Exit
Enter choice: 5
Enter Email to search: divija.puli@sjsu.edu
Student Found:
Email: divija.puli@sjsu.edu, Name: Divija Puli, Course: 1111, Grade: A, Marks: 100
Search Time: 0.000005 seconds
```

This functionality searches for a student using their email ID. It also measures and displays the time taken to perform the search using time.perf_counter(), satisfying the assignment requirement for performance measurement.

6. Sorting Students (WITH TIME)

a) Sort by Marks

```
Enter choice: 6

Sort By:
1. Marks Ascending
2. Marks Descending
3. Email Ascending
4. Email Descending
Enter choice: 1

Sorted Students:
Email: prad.mohan@sjsu.edu, Name: Prad Mohan, Course: 1111, Grade: C, Marks: 75
Email: Mady.dande@sjsu.edu, Name: Mady Dande, Course: 1111, Grade: C, Marks: 78
Email: vats.sri@sjsu.edu, Name: Vats sri, Course: 1111, Grade: B, Marks: 88
Email: sriya.anna@sjsu.edu, Name: Sriya Anna, Course: 1111, Grade: A, Marks: 98
Email: divija.puli@sjsu.edu, Name: Divija Puli, Course: 1111, Grade: A, Marks: 100

Sorting Time: 0.000258 seconds
```

```
Enter choice: 6

Sort By:
1. Marks Ascending
2. Marks Descending
3. Email Ascending
4. Email Descending
Enter choice: 2

Sorted Students:
Email: divija.puli@sjsu.edu, Name: Divija Puli, Course: 1111, Grade: A, Marks: 100
Email: sriya.anna@sjsu.edu, Name: Sriya Anna, Course: 1111, Grade: A, Marks: 98
Email: vats.sri@sjsu.edu, Name: Vats sri, Course: 1111, Grade: B, Marks: 88
Email: Mady.dande@sjsu.edu, Name: Mady Dande, Course: 1111, Grade: C, Marks: 78
Email: prad.mohan@sjsu.edu, Name: Prad Mohan, Course: 1111, Grade: C, Marks: 75

Sorting Time: 0.000046 seconds
```

This feature sorts student records based on marks in ascending or descending order. The sorting time is measured and displayed.

b) Sort by Email

```
Enter choice: 6

Sort By:
1. Marks Ascending
2. Marks Descending
3. Email Ascending
4. Email Descending
Enter choice: 3

Sorted Students:
Email: Mady.dande@sjsu.edu, Name: Mady Dande, Course: 1111, Grade: C, Marks: 78
Email: divija.puli@sjsu.edu, Name: Divija Puli, Course: 1111, Grade: A, Marks: 100
Email: prad.mohan@sjsu.edu, Name: Prad Mohan, Course: 1111, Grade: C, Marks: 75
Email: sriya.anna@sjsu.edu, Name: Sriya Anna, Course: 1111, Grade: A, Marks: 98
Email: vats.sri@sjsu.edu, Name: Vats sri, Course: 1111, Grade: B, Marks: 88

Sorting Time: 0.000027 seconds
```

```
Enter choice: 6

Sort By:
1. Marks Ascending
2. Marks Descending
3. Email Ascending
4. Email Descending
Enter choice: 4

Sorted Students:
Email: vats.sri@sjsu.edu, Name: Vats sri, Course: 1111, Grade: B, Marks: 88
Email: sriya.anna@sjsu.edu, Name: Sriya Anna, Course: 1111, Grade: A, Marks: 98
Email: prad.mohan@sjsu.edu, Name: Prad Mohan, Course: 1111, Grade: C, Marks: 75
Email: divija.puli@sjsu.edu, Name: Divija Puli, Course: 1111, Grade: A, Marks: 100
Email: Mady.dande@sjsu.edu, Name: Mady Dande, Course: 1111, Grade: C, Marks: 78

Sorting Time: 0.000026 seconds
```

This functionality sorts students alphabetically by email address. It demonstrates correct sorting logic and includes timing measurement.

7. Course Statistics

```
7. Course Statistics
8. Course Grade Report
9. Course Management
10. Professor Management
11. Login System
12. Exit
Enter choice: 7
Enter Course ID: 1111

Course ID: 1111
Number of Students: 5
Average Marks: 87.80
Median Marks: 88.00
```

This feature calculates the average and median marks for students within a specific course. It uses Python's built-in statistics functions to compute accurate results.

8. Course Report

```
8. Course Grade Report
9. Course Management
10. Professor Management
11. Login System
12. Exit
Enter choice: 8
Enter Course ID for report: 1111

--- Course Grade Report ---
Course ID: 1111
--------------------------------------------------
Divija Puli | Email: divija.puli@sjsu.edu | Marks: 100 | Grade: A
Sriya Anna | Email: sriya.anna@sjsu.edu | Marks: 98 | Grade: A
Vats sri | Email: vats.sri@sjsu.edu | Marks: 88 | Grade: B
Mady Dande | Email: Mady.dande@sjsu.edu | Marks: 78 | Grade: C
Prad Mohan | Email: prad.mohan@sjsu.edu | Marks: 75 | Grade: C
```

This report displays all students enrolled in a specific course along with their grades and marks. It helps analyze student performance per course.

9. Course CRUD Operations

a) Add Course

```
9. Course Management
10. Professor Management
11. Login System
12. Exit
Enter choice: 9

--- Course Management ---
1. Add Course
2. Display Courses
3. Delete Course
4. Update Course
5. Back
Enter choice: 1
Enter Course ID: 1112
Enter Course Name: Python for quitters
Enter Description: Programming for beginners and procrastinators
Course added successfully!
```

Adds a new course to courses.csv.

b) Display Course

```
--- Course Management ---
1. Add Course
2. Display Courses
3. Delete Course
4. Update Course
5. Back
Enter choice: 2

--- Courses ---
Course ID: 1111 | Name: Databases | Description: SQL and relational databases
Course ID: 1112 | Name: Python for quitters | Description: Programming for beginners and procrastinators
```

Displays all available courses in courses.csv

c) Delete Course

```
--- Course Management ---
1. Add Course
2. Display Courses
3. Delete Course
4. Update Course
5. Back
Enter choice: 3
Enter Course ID to delete: 1112
Course deleted successfully!
```

Removes a course from the system.

c) Update Course

```
--- Course Management ---
1. Add Course
2. Display Courses
3. Delete Course
4. Update Course
5. Back
Enter choice: 4
Enter Course ID to update: 1111
Leave blank to keep current value.
Course Name (Databases): Database and Sql
Description (SQL and relational databases):
Course updated successfully!
```

Updates course details.

10. Professor CRUD Operations

a) Add Professor

```
Enter choice: 10

--- Professor Management ---
1. Add Professor
2. Display Professors
3. Delete Professor
4. Update Professor
5. Professor Report
6. Back
Enter choice: 1
Enter Professor ID: param.saini@sjsu.edu
Enter Professor Name: Param
Enter Rank: Senior professor
Enter Course ID: 1112
Professor added successfully!
```

Adds a professor assigned to a course.

b) Display Professor

```
--- Professor Management ---
1. Add Professor
2. Display Professors
3. Delete Professor
4. Update Professor
5. Professor Report
6. Back
Enter choice: 2

--- Professors ---
Professor ID: yu.chen@sjsu.edu | Name: Yu Chen | Rank: Senior professor | Course ID: 1111
Professor ID: param.saini@sjsu.edu | Name: Param | Rank: Senior professor | Course ID: 1112
```

Displays all professors.

c) Update Professor

```
--- Professor Management ---
1. Add Professor
2. Display Professors
3. Delete Professor
4. Update Professor
5. Professor Report
6. Back
Enter choice: 4
Enter Professor ID to update: param.saini@sjsu.edu
Leave blank to keep current value.
Name (Param):
Rank (Senior professor):
Course ID (1112): 1114
Professor updated successfully!
```

Updates professor details.

d) Delete Professor

```
--- Professor Management ---
1. Add Professor
2. Display Professors
3. Delete Professor
4. Update Professor
5. Professor Report
6. Back
Enter choice: 3
Enter Professor ID to delete: param.saini@sjsu.edu
Professor deleted successfully!
```

Deletes professor record

.

11. Professor Report

```
Enter choice: 5
Enter Professor ID: yu.chen@sjsu.edu

--- Professor Report ---
Professor ID: yu.chen@sjsu.edu | Name: Yu Chen | Rank: Senior professor | Course ID: 1111

Students in their course:
-----------------------------------------------
Divija Puli | Marks: 100 | Grade: A
Sriya Anna | Marks: 98 | Grade: A
Vats sri | Marks: 88 | Grade: B
Mady Dande | Marks: 78 | Grade: C
Prad Mohan | Marks: 75 | Grade: C
```

This report shows all students taught by a specific professor based on course ID. It links professor data with student records.

12. Login System

a) Register User

```
--- Login System ---
1. Register
2. Login
3. Change Password
4. Back
Enter choice: 1
Enter Email (User ID): divimadhu126@gmail.com
Enter Password: ihatethiscommon
Enter Role (admin/student/professor): student
User registered successfully!
```

Stores user credentials with encrypted password.

login

User_id	Password	Role
divija.puli@sjsu.edu	tevzexmhizm	student
divimadhu126@sjsu.edu	lelele@55	student
divimadhu126@gmail.com	mlexixlmwgsqqsr	student

Encryption in login.csv

b) Login

```
--- Login System ---
1. Register
2. Login
3. Change Password
4. Back
Enter choice: 2
Enter Email: divimadhu126@gmail.com
Enter Password: ihatethiscommon
Login successful!
Logged in as student
```

Verifies users by decrypting stored passwords.

c) Change Password

```
--- Login System ---
1. Register
2. Login
3. Change Password
4. Back
Enter choice: 3
Enter Email: divimadhu126@gmail.com
Enter Current Password: ihatethiscommon
Enter New Password: imstressed
Password updated successfully!
```

This changes the password in login.csv and encrypts it again.

13. Unit Testing with execution time

```
(base) diviinity@Mac StudentManagementSystem_Lab1 % python -m unittest
..................
----------------------------------------------------------------------
Ran 18 tests in 0.048s

OK
```

Unit tests validate all functionalities including CRUD operations, searching, sorting, and data handling. The test cases generate and validate a dataset of 1000 student records using a temporary CSV file to ensure scalability and correctness

14. OOD Diagram

Attached in the Github repo as OOD_diagram.pdf

The diagram represents the system architecture including models, services, data handling, and relationships such as HAS-A associations between entities.